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Project Title:	Mechanotransduction Through Integrins		
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1. To compare the effects of mechanical stresses of different intensity, frequency and direction on focal adhesion assembly, cell strengthening and G protein activation
2. To analyze the Rho signaling mechanism responsible for cell strengthening behavior

3. To identify the molecules that mediate transmembrane mechanochemical coupling between integrins and G proteins within the focal adhesion

Task Description:

Over the past year, we showed that intracellular microtubules do bear large-scale compressive loads from a variety of physiological forces, but their buckling wavelength is reduced significantly because of mechanical coupling to the surrounding elastic cytoskeleton. We quantitatively explained this behavior, and showed that this coupling dramatically increases the compressive forces that microtubules can sustain, suggesting they can make a more significant structural contribution to the mechanical behavior of the cell than previously thought possible.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Elucidation of the mechanism by which integrins mediate cellular mechanotransduction should help us to understand the molecular basis of the cellular response to gravity as well as many other forms of mechanosensation and tissue regulation. Through the proposed research program, we hope to identify specific molecular mediators of mechanochemical transduction as well as novel mechanical signaling pathways in mammalian cells. These molecules may serve as potential targets for future modes of therapeutic intervention in diseases, such as atherosclerosis, hypertension, and osteoporosis, which are characterized by abnormal force transduction by cells. Further insight into mechanobiology also may impact on design of artificial tissues and medical devices for tissue repair and transplantation.

Task Progress:

AIM 1-
To better understand how cells sense and respond to changes in mechanical forces exerted on their extracellular matrix adhesions, we applied tensional forces to magnetic microbeads bound to cell surface integrin receptors using magnetic pulling cytometry and measured dynamic changes in bead displacement with sub-micrometer resolution using optical microscopy. These studies revealed that cells exhibit four distinct types of mechanical responses: 1) an immediate viscoelastic response to a brief 3 second force pulse that resulted in almost complete elastic recovery, 2) early adaptive behavior characterized by pulse-to-pulse attenuation in bead displacement in response to an oscillatory force regimen composed of multiple brief (3 second) force pulses separated by short (4 second) relaxation periods, 3) later adaptive cell stiffening in response to sustained (> 15 second) static stresses, and 4) a large-scale repositioning response in which beads exposed to prolonged (> 1 minute) static stresses were retracted back by the cell against the applied force. Importantly, these forms of cellular adaptation differed biochemically. Whereas the immediate viscoelastic response and early adaptation behavior were affected by dissipating cytoskeletal prestress (isometric tension) with inhibitors of either myosin ATPase or Rho-associated kinase (ROCK), the later adaptive response to longer (> 15 second) stresses was not sensitive to either of these inhibitors. The large-scale repositioning of beads in response to prolonged (> 1 minute) stress was prevented by inhibiting myosin-based tension generation through interference with Rho signaling, similar to the case of the immediate and early responses, but in addition it was prevented by blocking mechanosensitive ion channels using gadolinium or by inhibiting tyrosine kinases using PP2. All of these adaptive responses were suppressed by cooling the cells to 4°C to slow biochemical remodeling events, and the large-scale repositioning response required that integrins be chemically activated through occupancy of the RGD binding site. These results demonstrate that cells use multiple distinct mechanisms, including activation of small G proteins, to sense and respond to static and dynamic changes in the physical properties of the extracellular matrix that alter the level of mechanical stress applied to cell surface integrin receptors.

The formation of focal adhesions that mediate alterations of cell shape and movement is controlled by a mechanochemical mechanism in which cytoskeletal tensional forces drive changes in molecular assembly; however, little is known about the molecular biophysical basis of this response. In the past grant period, we developed a method to measure the unbinding rate constant of individual GFP-labeled focal adhesion molecules in living cells by modifying the fluorescence recovery after photobleaching (FRAP) technique and combining it with mathematical modeling. Using this method, we showed that decreasing cellular traction forces on focal adhesions by three different techniques – chemical inhibition of cytoskeletal tension generation, laser incision of an associated actin stress fiber, or use of compliant extracellular matrices – increases the of the focal adhesion protein zyxin. In contrast, the of another adhesion protein, vinculin, remains unchanged after tension dissipation. Mathematical models also demonstrated that these force-dependent increases in zyxin's that occur over seconds are sufficient to quantitatively predict large-scale focal adhesion disassembly that occurs physiologically over many minutes. These findings demonstrate that the molecular binding kinetics of some, but not all, focal adhesion proteins are sensitive to mechanical force, and suggest that force-dependent changes in this biophysical parameter may govern the supramolecular events that underlie focal adhesion remodeling in living cells.

Aim 2-

The small GTPase Rho plays a central role in integrin-dependent control of cell contractility, and has been shown to be activated by mechanical force application to surface integrin receptors. But the molecular basis of this mechanotransduction response remains unclear. To explore if mechanical force activates Rho downstream of integrin receptor activation, we plated endothelial cells on BioFlex membranes coated with a non-activating (K20) ß1 integrin antibody and Rho activation was measured by either the Rhotekin pull-down assay (Western blotting) or a novel FRET-based assay (confocal microscopy) after mechanical stretching. Application of mechanical force (15% strain) for 15 min significantly increased Rho activation in these cells compared to control cells. To determine the possible mechanism of Rho activation, we measured the translocation of p190Rho-GAP from the cytosol to the cytoskeleton fraction because this is an upstream regulator of Rho activity. We found that mechanical stretch increased the translocation of both p190RhoGAP and the tyrosine kinase, c-src, to the cytoskeletal fraction, and that this response paralleled the time course of Rho activation. To further confirm that mechanical force application activated Rho, we transfected these endothelial cells with a RhoFRET probe (Raich-RBD) which detects the changes in endogenous Rho activity. FRET efficiency was measured using the acceptor-photobleach method. Preliminary results suggest that application of 15% mechanical strain to cells adherent to fibronectin-coated BioFlex plates for 15 min induced a

statistically significant decrease in FRET efficiency which is indicative of Rho activation in response to mechanical force. Taken together, these results strongly suggest that mechanical force application through ß1 integrins results in stress-dependent activation of Rho.

Aim 3-

Mechanical stresses that are preferentially transmitted across the cell surface via transmembrane integrin receptors activate gene transcription by triggering production of intracellular chemical second messengers, such as cAMP. In the past grant period, we showed that the sensitivity of the cAMP signaling pathway to mechanical stresses transferred across ß1 integrins is mediated by force-dependent activation of some heterotrimeric G protein. During the grant funding period, we showed that Gas is recruited to focal adhesions that form within minutes following clustering of ß1 integrins induced by cell binding to magnetic microbeads coated with activating integrin ligands, and that ß1 integrin and Gas co-precipitate when analyzed biochemically. Stress application to activated ß1 integrins using magnetic twisting cytometry increases Gas recruitment and activates these large G proteins within focal adhesions, as measured by binding of biotinylated azido-anilido-GTP, whereas application of similar stresses to inactivated, integrins or control histocompatibility antigens has little effect. This response is relevant physiologically as application of mechanical strain to cells bound to flexible extracellular matrix-coated substrates induce translocation of phospho-CREB to the nucleus, which can be attenuated by inhibiting Gas activity, either using the inhibitor melittin or suppressing its expression using siRNA. Although integrins are not typical G protein-coupled receptors, these results show that integrins focus mechanical stresses locally on heterotrimeric G proteins within focal adhesions at the site of force application, and transduce mechanical stimuli into an intracellular cAMP signaling response by activating Gas at these membrane signaling sites.

Accurate understanding of cellular mechanotransduction through integrins and G proteins requires knowledge of how heterotrimeric G proteins associate with focal adhesions. Focal adhesions are large, complex and dynamic structures and the full complement of focal adhesion proteins are not known. Therefore, to determine the adaptor proteins bridging integrins to G proteins, we made a series of integrin mutants designed to disrupt large portions of the focal adhesion. In our last year of funding, we genetically engineered a series of six consecutive mutations covering the transmembrane and cytoplasmic domains of the ß1 integrin based on known interactions from the literature. These mutations are all made in chimeric constructs where the extracellular domain of integrin has been replaced with carbonic anhydrase IV (CAIV) epitope allowing us to selectively bind and study our mutants in the presence of wildtype integrins. We also generated two sets of integrin mutants, one with the ß1 transmembrane domain and another where the transmembrane domain has been replaced with the low density lipoprotein (LDL) transmembrane domain. Importantly, in preliminary studies using these mutants, we found that the cytoplasmic region of ß1 integrin is sufficient to support focal adhesion formation as determined by cell spreading and fluorescence microscopic analysis. Our preliminary studies indicate that, as expected, the various cytoplasmic mutants have differential spreading efficiencies. In addition, some have decreased focal adhesion strength in response to pulling through focal adhesions formed on CAIV antibody-coated beads despite their ability to spread on CAIV coated substrates indicating that the focal adhesion is partially disrupted. Therefore, these chimeras should be extremely useful tools for the dissection of mechanosensitive signal transduction through integrins and G proteins.

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